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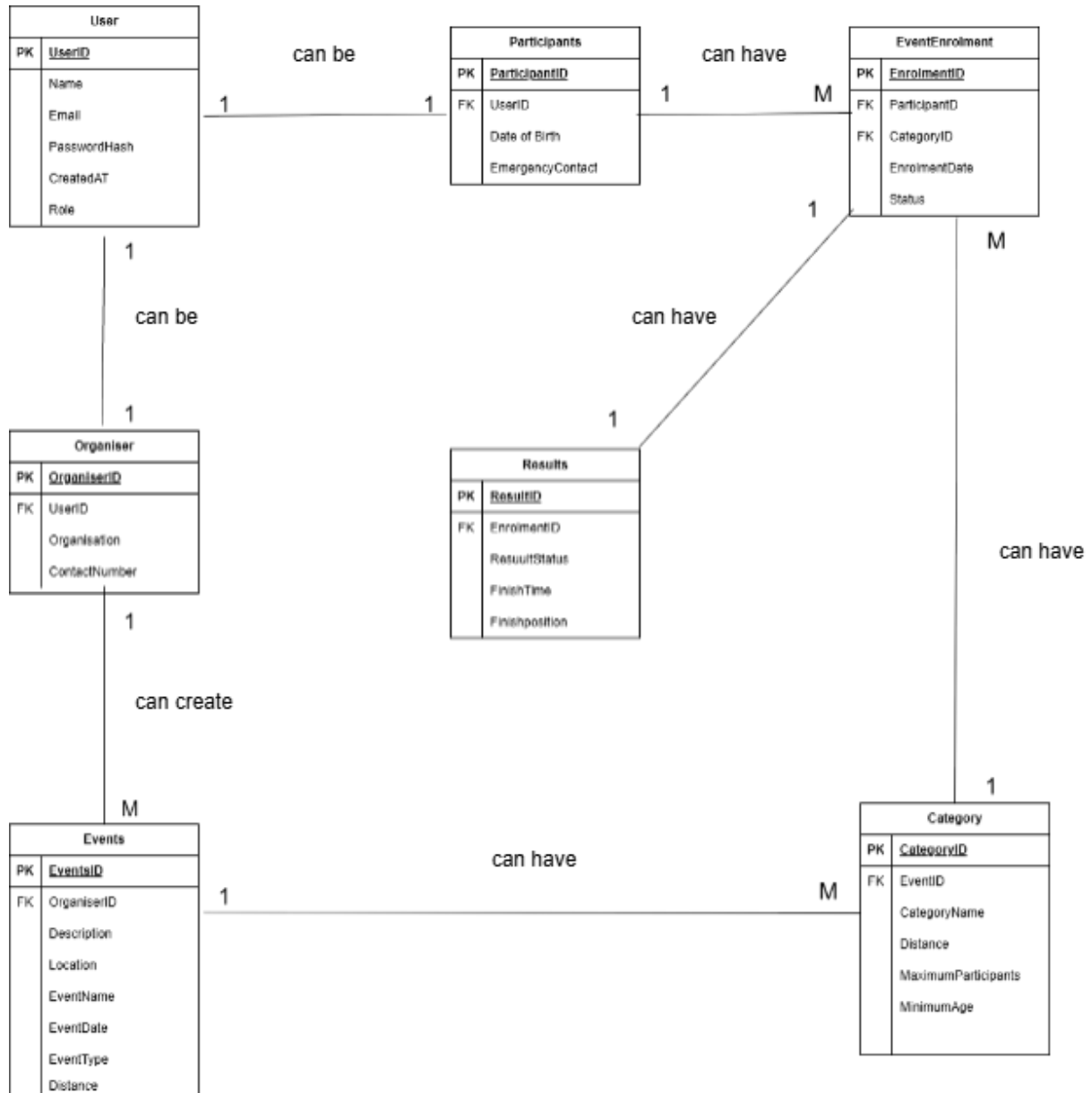
PROGRAMMING 2B

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SECTION A - ERD

ERD Raceday Management System



SECTION B - API ENDPOINT PLAN

API Endpoint Table

System: Raceday Event Management System

HTT P Met hod	Route	Descrip tion	Role Requir ed	Request Body	Expec ted Respo nse
POST	/api/auth/register	Register a new user account	Public	FullName,Email>Password Role	201 Created - User registered
POST	/api/auth/login	Authenticate a user and create a session	Public	Email, Password	200 OK - Login successful
GET	/api/users/me	View the currently logged-in user's profile	Authenticated	None	200 OK - profile details
PUT	/api/users/me	Update the currently logged-in user's profile	Authenticated	FullName,Email ContactDetails	200 OK - update profile
GET	/api/events	View all available RaceDay events	Public	None	200 OK - List of events
GET	/api/events/{id}	View details of a specific event	Public	None	200 OK - Event details

POST	/api/events	Create a new RaceDay event	Organiser	EventName, Description, EventDate, Location, Distance, EventType	201 Created - Event created
PUT	/api/events/{id}	Update an existing event	Organiser	EventName, Description, EventDate, Location, Distance, EventType	200 OK - event updated
DELETE	/api/events/{id}	Delete an existing event	Organiser	None	204 No Content - Event Deleted
GET	/api/events/{eventId}/categories	View categories available for an event	Public	None	200 OK - list of categories
POST	/api/events/{eventId}/categories	Create a category for an event	Organiser	CategoryName, DistanceKm, MaximumParticipants	201 Created - category created
PUT	/api/categories/{id}	Update an event category	Organiser	CategoryName, DistanceKm, MaximumParticipants	200 OK - category updated
DELETE	/api/categories/{id}	Delete an event category	Organiser	None	204 No Content - Category deleted
POST	/api/enrolments	Enrol a participant in an	Participant	CategoryID	201 Creat

		event category			ed-enrolment created
GET	/api/enrolments/me	View the participant's own enrolments	Participant	None	200 OK - List of enrolments
GET	/api/events/{eventId}/enrolments	View all participants enrolled in an organiser's event	Organiser	None	200 OK - list of enrolments
PUT	/api/enrolments/{id}/cancel	Cancel a participant's enrolment	Participant	None	200 OK - Enrolment cancelled
GET	/api/events/{eventId}/results	View results for an event	Public	None	200 OK - Event results
POST	/api/results	Record a participant's result	Organiser	EnrolmentID, FinishPosition, FinishTime, ResultStatus	201 Created - result recorded
PUT	/api/results/{id}	Update a participant's result	Organiser	FinishPosition, FinishTime, ResultStatus	200 OK - Results updated
GET	/api/participants/me/results	View the logged-in participant's results	Participant	None	200 OK - participant results

SECTION C - SQL DATABASE SCRIPT

System: RaceDay Event Management System

Database: RaceDay

CREATE DATABASE RaceDay;

USE RaceDay;

DROP TABLE IF EXISTS Results;

DROP TABLE IF EXISTS EventEnrolments;

DROP TABLE IF EXISTS Categories;

DROP TABLE IF EXISTS Events;

DROP TABLE IF EXISTS Participants;

DROP TABLE IF EXISTS Organisers;

DROP TABLE IF EXISTS Users;

-- Users Table

CREATE TABLE Users (

 UserID INT IDENTITY(1,1) PRIMARY KEY NOT NULL,

 FullName VARCHAR(100) NOT NULL,

 Email VARCHAR(100) NOT NULL UNIQUE,

 PasswordHash VARCHAR(255) NOT NULL,

 Role VARCHAR(20) NOT NULL,

 CHECK (Role IN ('Organiser', 'Participant'))

);

-- Organiser Table

CREATE TABLE Organisers (

 OrganiserID INT IDENTITY(1,1) PRIMARY KEY NOT NULL,

 UserID INT NOT NULL UNIQUE,

 OrganisationName VARCHAR(100) NOT NULL,

ContactNumber VARCHAR(20) NOT NULL,

FOREIGN KEY (UserID) REFERENCES Users(UserID)

);

-- Participant Table

CREATE TABLE Participants (

ParticipantID INT IDENTITY(1,1) PRIMARY KEY NOT NULL,

UserID INT NOT NULL UNIQUE,

DateOfBirth DATE NOT NULL,

EmergencyContact VARCHAR(100) NOT NULL,

FOREIGN KEY (UserID) REFERENCES Users(UserID)

);

-- Events Table

CREATE TABLE Events (

EventID INT IDENTITY(1,1) PRIMARY KEY NOT NULL,

OrganiserID INT NOT NULL,

EventName VARCHAR(150) NOT NULL,

Description VARCHAR(500) NOT NULL,

EventDate DATE NOT NULL,

Location VARCHAR(150) NOT NULL,

Distance DECIMAL(6,2) NOT NULL,

EventType VARCHAR(20) NOT NULL,

FOREIGN KEY (OrganiserID) REFERENCES Organisers(OrganiserID),

CHECK (Distance > 0),

CHECK (EventType IN ('Run', 'Walk', 'Cycle'))

);

-- Categories Table

```
CREATE TABLE Categories (  
    CategoryID INT IDENTITY(1,1) PRIMARY KEY NOT NULL,  
    EventID INT NOT NULL,  
    CategoryName VARCHAR(100) NOT NULL,  
    DistanceKm DECIMAL(6,2) NULL,  
    MinimumAge INT NULL,  
    MaximumAge INT NULL,  
    MaximumParticipants INT NOT NULL,  
    FOREIGN KEY (EventID) REFERENCES Events(EventID),  
    CHECK (DistanceKm IS NULL OR DistanceKm > 0),  
    CHECK (MaximumParticipants > 0)  
);
```

-- Event Enrolments Table

```
CREATE TABLE EventEnrolments (  
    EnrolmentID INT IDENTITY(1,1) PRIMARY KEY NOT NULL,  
    ParticipantID INT NOT NULL,  
    EventID INT NOT NULL,  
    CategoryID INT NOT NULL,  
    EnrolmentDate DATETIME NOT NULL DEFAULT GETDATE(),  
    Status VARCHAR(20) NOT NULL DEFAULT 'Registered',  
    FOREIGN KEY (ParticipantID) REFERENCES Participants(ParticipantID),  
    FOREIGN KEY (EventID) REFERENCES Events(EventID),  
    FOREIGN KEY (CategoryID) REFERENCES Categories(CategoryID),  
    CHECK (Status IN ('Registered', 'Cancelled', 'Completed'))  
);
```

-- Results Table

CREATE TABLE Results (

ResultID INT IDENTITY(1,1) PRIMARY KEY NOT NULL,

EnrolmentID INT NOT NULL UNIQUE,

FinishPosition INT NOT NULL,

FinishTime TIME NOT NULL,

ResultStatus VARCHAR(20) NOT NULL DEFAULT 'Finished',

FOREIGN KEY (EnrolmentID) REFERENCES EventEnrolments(EnrolmentID),

CHECK (FinishPosition > 0),

CHECK (ResultStatus IN ('Finished', 'Disqualified', 'Did Not Finish'))

);

-- Insert Users

INSERT INTO Users (FullName, Email, PasswordHash, Role)

VALUES

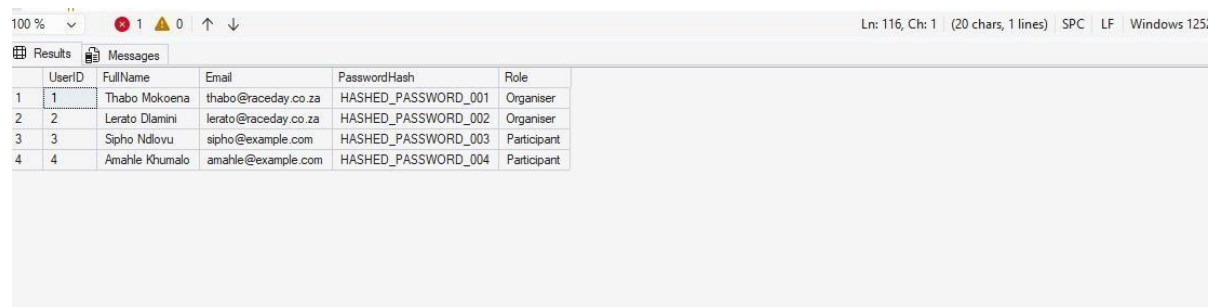
('Thabo Mokoena', 'thabo@raceday.co.za', 'HASHED_PASSWORD_001', 'Organiser'),

('Lerato Dlamini', 'lerato@raceday.co.za', 'HASHED_PASSWORD_002', 'Organiser'),

('Sipho Ndlovu', 'sipho@example.com', 'HASHED_PASSWORD_003', 'Participant'),

('Amahle Khumalo', 'amahle@example.com', 'HASHED_PASSWORD_004', 'Participant');

SELECT * FROM Users;



	UserID	FullName	Email	PasswordHash	Role
1	1	Thabo Mokoena	thabo@raceday.co.za	HASHED_PASSWORD_001	Organiser
2	2	Lerato Dlamini	lerato@raceday.co.za	HASHED_PASSWORD_002	Organiser
3	3	Sipho Ndlovu	sipho@example.com	HASHED_PASSWORD_003	Participant
4	4	Amahle Khumalo	amahle@example.com	HASHED_PASSWORD_004	Participant

-- Insert Organisers

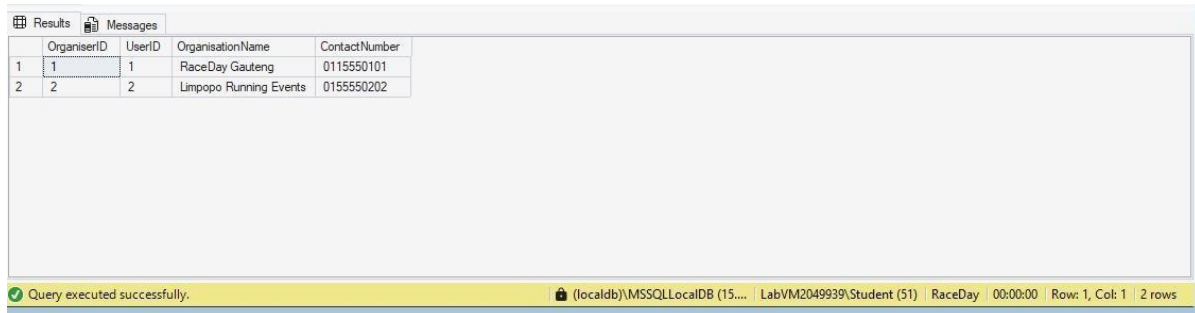
INSERT INTO Organisers (UserID, OrganisationName, ContactNumber)

VALUES

(1, 'RaceDay Gauteng', '0115550101'),

(2, 'Limpopo Running Events', '0155550202');

SELECT * FROM Organisers;



OrganiserID	UserID	OrganisationName	ContactNumber
1	1	RaceDay Gauteng	0115550101
2	2	Limpopo Running Events	0155550202

Query executed successfully. (localdb)\MSSQLLocalDB (15... | LabVM2049939\Student (51) | RaceDay | 00:00:00 | Row: 1, Col: 1 | 2 rows

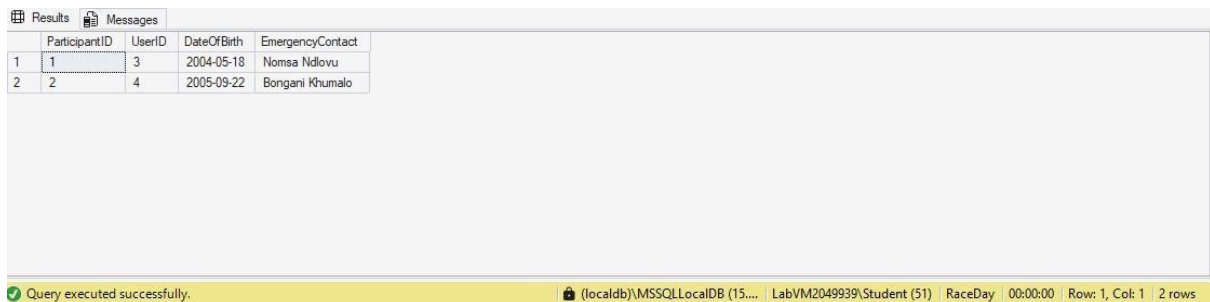
INSERT INTO Participants (UserID, DateOfBirth, EmergencyContact)

VALUES

(3, '2004-05-18', 'Nomsa Ndlovu'),

(4, '2005-09-22', 'Bongani Khumalo');

SELECT * FROM Participants;



ParticipantID	UserID	DateOfBirth	EmergencyContact
1	3	2004-05-18	Nomsa Ndlovu
2	4	2005-09-22	Bongani Khumalo

Query executed successfully. (localdb)\MSSQLLocalDB (15... | LabVM2049939\Student (51) | RaceDay | 00:00:00 | Row: 1, Col: 1 | 2 rows

INSERT INTO Events

(OrganiserID, EventName, Description, EventDate, Location, Distance, EventType)

VALUES

(1, 'Johannesburg City Run', 'Annual city running event', '2026-10-10', 'Johannesburg', 10.00,

'Run'),

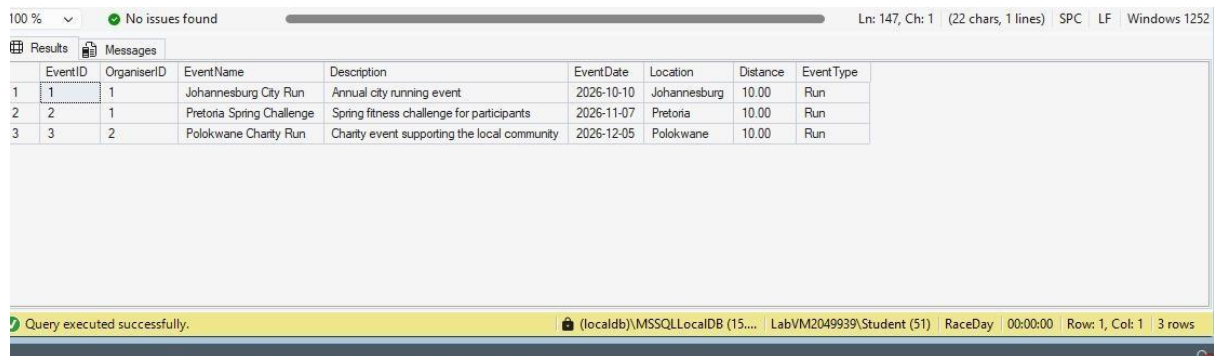
(1, 'Pretoria Spring Challenge', 'Spring fitness challenge for participants', '2026-11-07', 'Pretoria', 10.00,

'Run'),

(2, 'Polokwane Charity Run', 'Charity event supporting the local community', '2026-12-05', 'Polokwane', 10.00,

'Run');

SELECT * FROM Events;



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	EventID	OrganiserID	EventName	Description	EventDate	Location	Distance	EventType
1	1	1	Johannesburg City Run	Annual city running event	2026-10-10	Johannesburg	10.00	Run
2	2	1	Pretoria Spring Challenge	Spring fitness challenge for participants	2026-11-07	Pretoria	10.00	Run
3	3	2	Polokwane Charity Run	Charity event supporting the local community	2026-12-05	Polokwane	10.00	Run

Query executed successfully. (localdb)\MSSQLLocalDB (15... LabVM2049939\Student (51) RaceDay 00:00:00 Row: 1, Col: 1 3 rows

-- Insert Categories

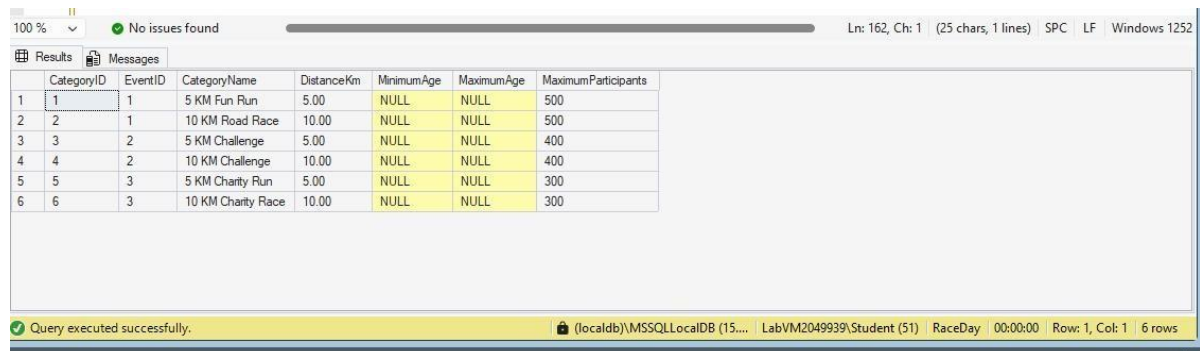
INSERT INTO Categories

(EventID, CategoryName, DistanceKm, MinimumAge, MaximumAge, MaximumParticipants)

VALUES

(1, '5 KM Fun Run', 5.00, NULL, NULL, 500),
(1, '10 KM Road Race', 10.00, NULL, NULL, 500),
(2, '5 KM Challenge', 5.00, NULL, NULL, 400),
(2, '10 KM Challenge', 10.00, NULL, NULL, 400),
(3, '5 KM Charity Run', 5.00, NULL, NULL, 300),
(3, '10 KM Charity Race', 10.00, NULL, NULL, 300);

SELECT * FROM Categories;



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	CategoryID	EventID	CategoryName	DistanceKm	MinimumAge	MaximumAge	MaximumParticipants
1	1	1	5 KM Fun Run	5.00	NULL	NULL	500
2	2	1	10 KM Road Race	10.00	NULL	NULL	500
3	3	2	5 KM Challenge	5.00	NULL	NULL	400
4	4	2	10 KM Challenge	10.00	NULL	NULL	400
5	5	3	5 KM Charity Run	5.00	NULL	NULL	300
6	6	3	10 KM Charity Race	10.00	NULL	NULL	300

Query executed successfully. (localdb)\MSSQLLocalDB (15... LabVM2049939\Student (51) RaceDay 00:00:00 Row: 1, Col: 1 6 rows

INSERT INTO EventEnrolments

(ParticipantID, EventID, CategoryID, Status)

VALUES

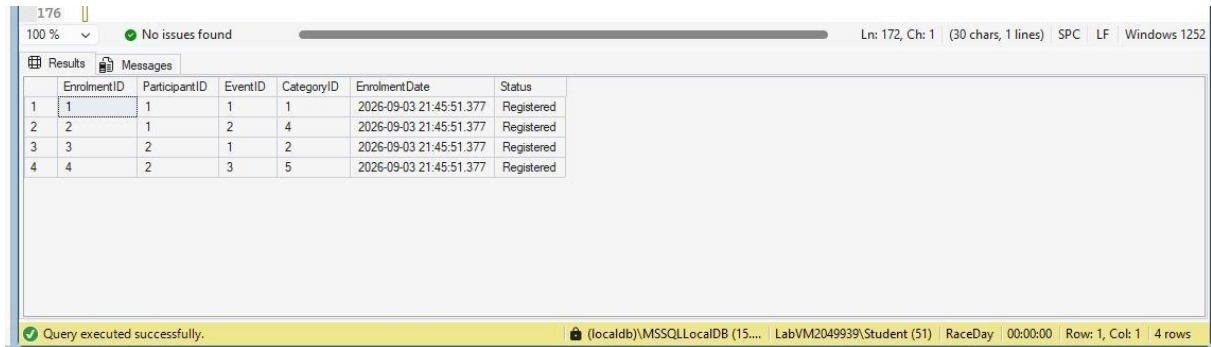
(1, 1, 1, 'Registered'),

(1, 2, 4, 'Registered'),

(2, 1, 2, 'Registered'),

(2, 3, 5, 'Registered');

SELECT * FROM EventEnrolments;



176 | 100 % | No issues found | Ln: 172, Ch: 1 | (30 chars, 1 lines) | SPC | LF | Windows 1252

	EnrolmentID	ParticipantID	EventID	CategoryID	EnrolmentDate	Status
1	1	1	1	1	2026-09-03 21:45:51.377	Registered
2	2	1	2	4	2026-09-03 21:45:51.377	Registered
3	3	2	1	2	2026-09-03 21:45:51.377	Registered
4	4	2	3	5	2026-09-03 21:45:51.377	Registered

Query executed successfully. | (localdb)\MSSQLLocalDB (15.... | LabVM2049939\Student (51) | RaceDay | 00:00:00 | Row: 1, Col: 1 | 4 rows

-- Insert Results

INSERT INTO Results

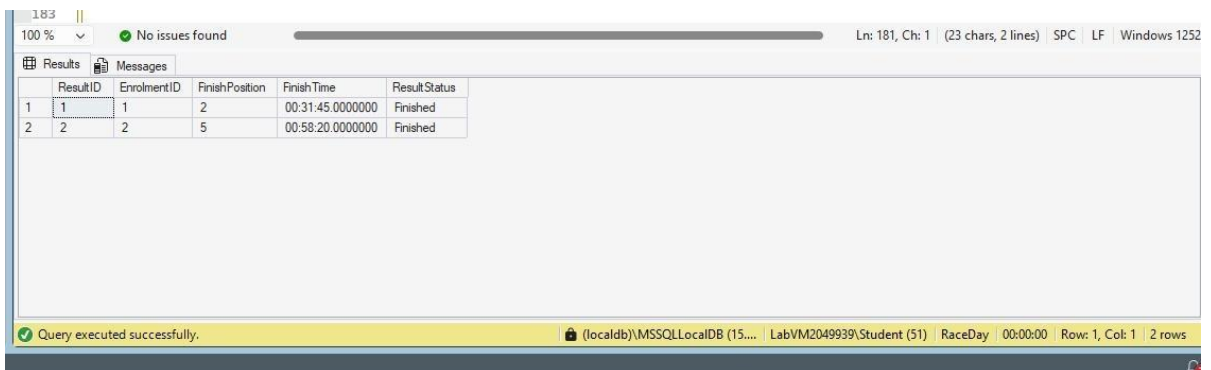
(EnrolmentID, FinishPosition, FinishTime, ResultStatus)

VALUES

(1, 2, '00:31:45', 'Finished'),

(2, 5, '00:58:20', 'Finished');

SELECT * FROM Results;



183 | 100 % | No issues found | Ln: 181, Ch: 1 | (23 chars, 2 lines) | SPC | LF | Windows 1252

	ResultID	EnrolmentID	FinishPosition	FinishTime	ResultStatus
1	1	1	2	00:31:45.0000000	Finished
2	2	2	5	00:58:20.0000000	Finished

Query executed successfully. | (localdb)\MSSQLLocalDB (15.... | LabVM2049939\Student (51) | RaceDay | 00:00:00 | Row: 1, Col: 1 | 2 rows

REFERENCES

Microsoft (2024) *CREATE TABLE (Transact-SQL)*. Microsoft Learn. Available at: Microsoft Learn - CREATE TABLE (Transact-SQL) (Accessed: 3 September 2026).

Microsoft (2024) *Primary and foreign key constraints*. Microsoft Learn. Available at: Microsoft Learn - Primary and foreign key constraints (Accessed: 3 September 2026).

Coronel, C. and Morris, S. (201G) *Database Systems: Design, Implementation, & Management*. 13th edn. Boston: Cengage Learning.